



BGP TROUBLESHOOTING FLOWCHART

STEP-BY-STEP PATH TO ROOT CAUSE

LEGEND

- Reachability (L3)
- Transport (TCP)
- BGP Process
- Analysis / Verify
- Action / Fix



START

BGP Session / Routes Not Working



STEP 1

REACHABILITY
Layer 3



1. CAN YOU REACH THE PEER?

ping <peer-IP> source <loopback>
(Use the source IP configured for BGP)

NO

NOT REACHABLE

- Check IGP / Static route
- Verify VPN / VRF
- Check IP address / mask
- Check firewall / ACL

Fix & Retest

YES



STEP 2

TRANSPORT
TCP / Port 179



2. IS TCP 179 OPEN?

telnet <peer-IP> 179
or show tcp brief | include 179

NO

TCP 179 NOT OPEN

- Firewall / ACL blocking
- Wrong source IP
- Routing asymmetry
- GTSM / TTL security issue
- Both sides passive

Fix & Retest

YES



STEP 3

BGP STATE
Identifying Issue



3. WHAT IS THE BGP STATE?

show bgp neighbors <peer-IP>

IDLE

Not attempting
(No TCP)



CONNECT

TCP SYN sent
(No SYN-ACK)



ACTIVE

TCP failing
(Retrying)



OPENSENT

OPEN sent
(No response)



OPENCONFIRM

OPEN+KEEPALIVE sent
(Awaiting KEEPALIVE)



ESTABLISHED

Session up
(But no routes?)



STEP 4

DEEP DIVE
State-Specific
Analysis

WHY?

- No route to peer
- Neighbor shut down
- Wrong VRF
- Hold-down timer
- Passive mode

CHECK:

- show ip route
- show bgp neighbors
- show run | s bgp
- show bgp nei <IP>

WHY?

- Port 179 blocked
- Source IP mismatch
- Routing asymmetry
- GTSM / TTL issue
- Both sides passive

CHECK:

- telnet <IP> 179
- show tcp brief
- show bgp neighbors
- show access-lists

WHY?

- Source IP unreachable
- eBGP multihop not set
- TTL issue (GTSM)
- ACL dropping return
- MTU / PMTUD issue
- Wrong neighbor IP

CHECK:

- ping <IP> source Lo0
- show bgp neighbors
- show ip interface Lo0
- show access-lists

WHY?

- Wrong remote AS
- BGP version mismatch
- Router-ID collision
- Hold time too low
- AF capability mismatch

CHECK:

- show bgp neighbors
- debug bgp events
- show logging | i BGP
- packet capture 179

WHY?

- MD5 auth mismatch
- Hold timer expired
- GTSM mismatch
- Excessive CPU / load

CHECK:

- show bgp neighbors
- show logging | i BGP
- debug bgp events
- show processes cpu

WHY?

- No network statement
- Route-map filtering
- Next-hop unreachable
- Max-prefix limit hit
- Policy not exporting

CHECK:

- show bgp summary
- show ip bgp
- show bgp neighbors
- show ip route bgp



STEP 5

ANALYZE
Logs / Notifications
Capture



5. CHECK LOGS, NOTIFICATIONS & CAPTURE (IF NEEDED)

CHECK LOGS

- show logging | i BGP
- show bgp neighbors <IP>
- Look for NOTIFICATION messages

COMMON NOTIFICATION CODES

- 2.2 - Bad Peer AS
- 2.3 - Bad BGP Identifier
- 2.7 - Unsupported Capability
- 4.2 - Hold Timer Expired
- 6.1 - Maximum Number of Prefixes
- 6.5 - Administrative Shutdown

PACKET CAPTURE

- capture / monitor tcp port 179
- Verify OPEN, KEEPALIVE, NOTIFICATION messages
- Check TCP flags / resets



APPLY FIX
& VALIDATE



6. APPLY FIX & RESET SESSION (SOFT)

APPLY CORRECT CONFIGURATION

- Update neighbor config
- Fix routes / ACL / Firewall
- Set update-source / multihop
- Match MD5 / TTL security
- Adjust hold timers

RESET (SOFT RESET RECOMMENDED)

Cisco IOS / IOS-XE / NX-OS
clear ip bgp <peer-IP> soft in
clear ip bgp <peer-IP> soft out
JUNIPER
clear bgp neighbor <peer-IP> soft in
clear bgp neighbor <peer-IP> soft out

WAIT & MONITOR

- Give the session time
- Monitor state transitions
- Check for flaps / errors



STEP 7
ROUTES



7. VERIFY ROUTES & END STATE

VERIFY PREFIXES

- show bgp summary
- Check prefix count
- show ip bgp | include <prefix>
- show bgp neighbors <IP> received-routes

VERIFY REACHABILITY

- show ip route <prefix>
- traceroute <prefix> source <loopback>
- Test traffic flow end-to-end



SUCCESS!
ESTABLISHED
& ROUTES FLOWING



BGP ISP TROUBLESHOOTING GUIDE

From Idle to Established — and Everything That Breaks In Between

any error → hold-down timer



1 IDLE

- WHY:**
- neighbor not configured or shutdown command active
 - No route to peer IP in routing table (check IGP/static)
 - BGP router process not running (missing router bgp <ASN>)
 - Hold-down timer from previous error — wait or clear
 - Wrong VRF — neighbor defined in default, peering in VRF (or vice versa)
 - Passive mode: neighbor <IP> passive — will not initiate TCP

FIX:

```
show bgp neighbors <IP> | i "BGP state|shutdown"
show route <peer-IP> / show ip route <peer-IP>
clear bgp <neighbor> [IOS: clear ip bgp <IP>]
Juniper: show bgp neighbor <IP> | match "Type|State"
```

2 CONNECT

- WHY:**
- Firewall or ACL blocking TCP port 179 inbound OR the TCP return path
 - Source IP mismatch — router sending from physical IP, ISP expects loopback
 - Routing asymmetry — SYN leaves, TCP SYN-ACK dropped on return
 - GTSM/TTL Security enabled on one side only (ttl-security hops mismatch)
 - Both sides configured passive — neither initiates

FIX:

```
telnet <peer-IP> 179 from the router CLI — proves port 179 is open
show bgp neighbors <IP> | i "Local host|Foreign host"
show tcp brief — confirm whether TCP session exists at all
Verify ACL/firewall permits TCP 179 both directions
Juniper: run telnet <IP> port 179
```

3 ACTIVE

- WHY:**
- update-source set to loopback but loopback not in IGP / not reachable
 - eBGP multihop not configured — default TTL=1 drops packet at first hop
 - Wrong neighbor IP configured (off-by-one in /30 — classic typo)
 - ACL silently dropping TCP return traffic (not logging, no counter increment)
 - GTSM mismatch — one side has ttl-security hops, other sends TTL=255
 - MTU mismatch — TCP SYN gets through, large BGP OPEN fragment drops
 - disable-connected-check missing for loopback peering on directly connected link

FIX:

```
ping <peer-IP> source <loopback> — must succeed before BGP will work
show bgp neighbors <IP> | i "BGP state|Local host|Remote"
neighbor <IP> ebgp-multihop 2 (if peering via loopback)
neighbor <IP> update-source Loopback0 (if peering via loopback)
neighbor <IP> disable-connected-check (loopback on directly connected link)
Juniper: set protocols bgp group <name> multihop ttl <n>
```

4 OPENSENT

- WHY:**
- Wrong remote AS configured — BGP rejects OPEN (NOTIFICATION code 2.2)
 - 2-byte AS configured locally, ISP using 4-byte AS (or vice versa)
 - Router-ID collision — both peers accidentally using same BGP router-ID
 - Address-family capability mismatch (e.g. IPv6 AF not accepted)
 - Hold time < 3 seconds rejected per RFC 4271 — code 2.6
 - Remote end receives OPEN, sends back NOTIFICATION, resets session

FIX:

```
Confirm ASN with ISP — 2-byte (< 65536) or 4-byte (> 65535)
show bgp neighbors <IP> | i "BGP version|remote AS|BGP Identifier"
Check syslog/log buffer for NOTIFICATION messages with error codes
debug bgp <neighbor> events (IOS: debug ip bgp <IP> events)
Packet capture port 179 — read the OPEN message fields directly
Juniper: monitor traffic interface <int> matching "tcp port 179"
```

5 OPENCONFIRM

- WHY:**
- MD5 authentication password mismatch (one wrong character)
 - Hold timer expired before KEEPALIVE received
 - GTSM/TTL Security mismatch at this stage
 - Excessive CPU load — router can't respond with KEEPALIVE in time
 - TCP window size / buffer exhaustion under heavy load

FIX:

```
Remove MD5 auth temporarily to test (NO AUTH = TEST ONLY)
confirm password case-sensitive, no extra spaces
show bgp neighbors <IP> | i authentication
show bgp neighbors <IP> | i holdtime
increase hold-time if needed (recommend 60-180 sec)
Juniper: set protocols bgp group <name> hold-time <secs>
```

6 ESTABLISHED

- WHY:**
- Session up but no routes? Likely policy / reachability problem
 - Route filtering — prefix-lists / route-maps / communities
 - Next-hop not reachable — traffic blackholed even though routes received
 - Network statement missing or incorrect on your side
 - Max-prefix limit hit — routes received but not installed
 - Outbound routes suppressed by policy

FIX:

```
show bgp summary — check prefix counts (sent/received)
show bgp neighbors <IP> advertised-routes
show bgp neighbors <IP> received-routes
show ip route bgp <prefix> — verify next-hop reachable?
traceroute <prefix> source <loopback>
Juniper: show route receive-protocol bgp <group>
```

VERIFICATION COMMANDS — Run In This Order

- | | |
|--|---|
| 1. show bgp summary | → all neighbors + state + prefix count |
| 2. show bgp neighbors <IP> | → full state detail + error codes |
| 3. show bgp neighbors <IP> advertised-routes | → what you're actually sending |
| 4. show bgp neighbors <IP> received-routes | → what ISP is sending |
| 5. show ip route bgp | → routes installed in RIB |
| 5. ping <peer> source <loopback> | → verify Layer 3 reachability first |
| 7. telnet <peer> 179 | → test TCP port 179 directly |
| 8. clear ip bgp <IP> soft | → refresh routes without dropping session |

NOTE: Command syntax and output may vary between different vendors (Cisco, Juniper, Arista, etc.)

★ GOLDEN RULES

- | | |
|---|---|
| ✓ Test TCP 179 first — BGP won't come up if it's blocked | ✓ Never clear bgp * in production without a maintenance window |
| ✓ ACTIVE state = Layer 3 or source IP problem, almost always | ✓ Hold timer mismatch (Cisco 180s vs Juniper 90s) causes session flapping |
| ✓ MD5 failures are silent — one wrong char = same symptoms as firewall drop | ✓ Full BGP table ≈ 900K routes — verify RAM before accepting full table |
| ✓ Established ≠ traffic working — verify prefix counts and next-hop | |